

Module Geomodelling for Natural Resource Evaluation

Exercise

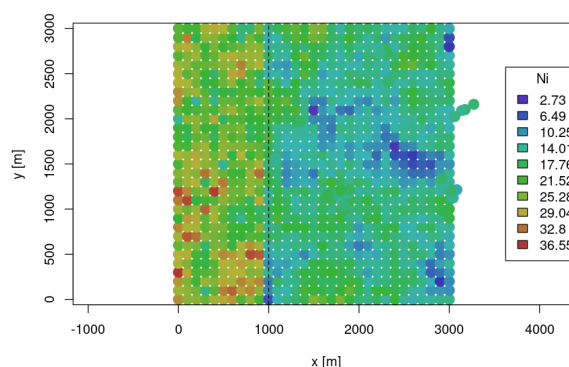
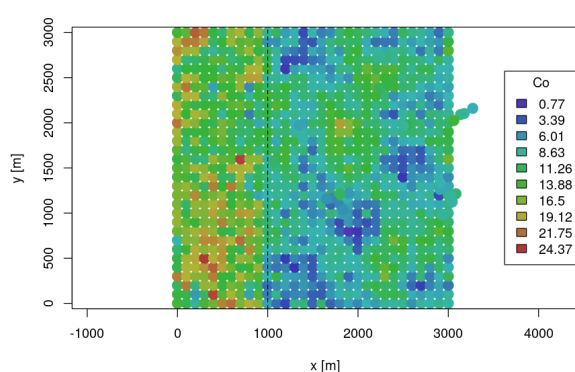
Summer Semester 2026

Variogram analysis

Do a variogram analysis for the testData123.csv data set and send me a short pdf document with your findings and plots and the R-code which you needed to answer the questions until 14 June.

Please compare the variogram of both parts of the area with the one of the complete area.

Do they have the same variogram or would you use a different variogram? Give the parameter of all three variograms in a table.



During the variogram analysis you should do the following steps and try to answer the questions in your report:

- look at the variogram cloud, does anything looks strange, up to which distances we have many pairs, are there outliers/ outlier bands, extrem points
- empirical variogram - try to guess up to which distance you can interpret it
- which variogrammodel would fit? Does a simple model fit or do we need a nested model? give a rough estimat of nugget, sill and range
- fit a varigram model
- check the fit
- do you have enough points (visual check / 30 - 50 pairs in the groups)?
- draw your variogram model
- if it doesn't fit, decide on directional range and sill
- check for directional dependency - draw a variogram map
- if it looks like directional dependency find main angles and draw directional variograms
- do you have enough points (30 - 50 pairs in the groups)?
- prepaire a directional variogram model, fit it, check it

Deadline: June 14, 2026

Upload the pdf file of your report and the R-Script in the Opal-course in the folder „Hand In Exercise Variogram“